CSE299 Week 7 Project Update Report

Project Title: Dhaka Air Quality Prediction System
Course: CSE299 **Section:** 14 **Group:** 8

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1. Week 7 Objective

The main objective of Week 7 was to begin integrating the machine-learning work completed in Week 6 with a working web application prototype. The focus was not to complete the final dashboard, but to create an initial system where users can provide air-quality and meteorological information and receive a next-hour PM2.5 prediction from the trained model.

2. Work Completed

During Week 7, we developed the first working prototype of the **Dhaka Air Quality Predictor** web application.

- Developed a dark-themed web dashboard for the project.
- Added monitoring-station selection for BARC/Farmgate, Darussalam, and DoE/Agargaon.
- Added input fields for pollution and weather-related variables.
- Connected the Week 6 Random Forest forecasting model to a Python backend.
- Added PM2.5 lag features for 1-hour, 3-hour, 6-hour, and 24-hour historical observations.
- Configured the model to predict PM2.5 concentration one hour ahead.
- Added prediction output and air-quality interpretation to the dashboard.
- Added model-performance information and feature-importance visualization.
- Added an actual-vs-predicted visualization for evaluating forecasting performance.

3. Model Integration

The Random Forest model developed during Week 6 is now connected to the web application backend.

Workflow: User Input → Backend Processing → Time-Series Features → Random Forest Model → Next-Hour PM2.5 Prediction → Dashboard Output

The model uses historical PM2.5 values together with available pollution, weather, time, and monitoring-station information. Important time-series features include current PM2.5 and PM2.5 from 1, 3, 6, and 24 hours earlier.

4. Model Performance

The Random Forest model selected during Week 6 achieved the following results on the 2024+ test data:

Metric	Result
RMSE	32.36
R^2	0.788
MAE	16.18

The Week 7 prototype uses this trained model for its prediction workflow. The dashboard presents predicted PM2.5 concentration as the main ML output.

5. Web Application Prototype

- Dataset information and monitoring-station selection
- Pollution and meteorological input controls
- Next-hour PM2.5 prediction and air-quality interpretation
- Model status and model-performance information
- Feature-importance and actual-versus-predicted visualizations

The interface is currently an **initial prototype**. Full dashboard development and further integration will continue in later project stages.

6. Improvements from Week 6

Week 6: Time-series forecasting → lag features → multiple-model comparison → error analysis.

Week 7: Trained ML Model → Python Backend → Web Interface → Prediction Output.

Week 7 represents the first integration between the machine-learning component and the user-facing system.

7. Current Limitations

- The dashboard is not yet the final production version.
- Additional testing is required with different input combinations.
- The prediction workflow needs further validation before final deployment.
- More dashboard features and data visualization will be added later.
- The model predicts next-hour PM2.5 rather than directly predicting official AQI.

8. Next Steps

- Improve and test the web application.
- Continue analyzing forecasting errors and improve useful time-series features.
- Finalize the model selected for the application.
- Improve graphs, result visualization, and dashboard functionality.
- Prepare the system for final integration and demonstration.

9. Week 7 Status

Week 7 milestone: Initial web prototype and ML model integration completed.

The project now has a working path from cleaned historical air-quality data to time-series model prediction and an initial web-based user interface.